Analysis of Factors Associated with Poor Response to Nutritional Treatment in Severely Malnourished Children Aged Under 6 Months.

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1. Methodology

Descriptive analyses were conducted to characterize the cohort and the non-response to the nutritional treatment. Continuous variables were summarized using medians and interquartile ranges (25th - 75th percentile) due to their non-normal distribution, whereas categorical variables were summarized as counts and percentages (%).

Differences between groups (success and failure) were assessed using the Wilcoxon rank-sum test or Kruskal–Wallis test for continuous variables and Pearson's chi-square or Fisher's exact test for categorical variables, as appropriate.

Univariable logistic regression analyses were performed to estimate crude associations between independent variables and the dependent variable. Variables considered clinically relevant or statistically associated with the non-response to the nutritional treatment at $\underline{p} < 0.20$ were included in a multivariable logistic regression.

Prior to model building, pairwise correlations among independent variables were examined to identify highly correlated predictors and assess potential multicollinearity. Pearson correlation coefficients were computed for continuous variables (or Spearman's rank correlations for non-normally distributed variables). Variables exhibiting strong correlations ($|r| \geq 0.70$) were carefully evaluated to avoid redundancy. Variance Inflation Factors (VIFs) were subsequently calculated in the multivariable model, with VIF values > 5 indicating potential multicolliniarity.

Model selection was guided by likelihood ratio tests and the principle of parsimony. A stepwise selection procedure was used to identify the most parsimonious model (based on the Akaike Information Criterion (AIC)) by retaining variables that improved model fit while excluding predictors that contributed little additional information.

Adjusted odds ratios (aORs) with 95% confidence intervals (CIs) were reported. Statistical significance in the final model was defined as a two-sided $\underline{p} < 0.05$. A variable was significantly associated with the risk of nutritional treatment failure if: i) the odds ratio (OR) was > 1 with a lower limit of the confidence interval (CI) > 1; or ii) OR<1 with upper CI limit <1; and p-value was <0.05.

All analyses were conducted using R (version 4.3.0).

2. Study population

Insert comments here.

3. Descriptive Characteristics

Insert comments here.

4. Bivariate Analysis

Table 1: Association between explanatory variables and outcome at phase 1.

	Non-response to treatment in Phase				
Characteristic	**FALSE** N = 221[1]	**TRUE** N = 32[1]	**Overall** N = 253[1]	**p-value**[23]	**q-value**[4]
Age (Month)				0.135	0.344
Median (Q1 ; Q3)	2 (1.0 ; 4)	3 (2.0 ; 4)	3 (1.0 ; 4)		
Min ; Max	0 ; 10.0	0 ; 9.0	0 ; 10.0		
Age group				**0.015***	0.084
[0;6[	213 (96.4%)	27 (84.4%)	240 (94.9%)		
[6;25[	8 (3.6%)	5 (15.6%)	13 (5.1%)		
Gender				0.062	0.203
Male	113 (51.1%)	22 (68.8%)	135 (53.4%)		
Female	108 (48.9%)	10 (31.3%)	118 (46.6%)		
Locality				0.393	0.521
Urban	93 (43.5%)	16 (51.6%)	109 (44.5%)		
Rural	121 (56.5%)	15 (48.4%)	136 (55.5%)		
Missing values	7	1	8		
Vaccination status				0.128	0.344
Up to date	50 (61.7%)	12 (85.7%)	62 (65.3%)		
Out of date	31 (38.3%)	2 (14.3%)	33 (34.7%)		
Missing values	140	18	158		
Breastfeeding status					
Yes	221 (100.0%)	32 (100.0%)	253 (100.0%)		
Type admission				0.337	0.521
New admission	191 (90.5%)	26 (83.9%)	217 (89.7%)		
Readmission	20 (9.5%)	5 (16.1%)	25 (10.3%)		
Missing values	10	1	11		
Morbidity burden				0.360	0.521

Characteristic	Non-response to treatment in Phase				
	FALSE N = 221[1]	**TRUE** N = 32[1]	**Overall** N = 253[1]	**p-value**[23]	**q-value**[4]
Median (Q1 ; Q3)	1 (0.0 ; 2)	2 (0.0 ; 3)	1 (0.0 ; 2)		
Min ; Max	0 ; 5.0	0 ; 6.0	0 ; 6.0		
Shock	1 (0.5%)	0 (0.0%)	1 (0.4%)	>0.999	>0.999
Missing values	24	4	28		
Gastroenteritis	116 (58.9%)	20 (71.4%)	136 (60.4%)	0.204	0.391
Missing values	24	4	28		
Pneumonia	103 (52.3%)	14 (50.0%)	117 (52.0%)	0.821	0.944
Missing values	24	4	28		
Tuberculosis	2 (1.0%)	3 (10.7%)	5 (2.2%)	**0.015***	0.084
Missing values	24	4	28		
Digestive candidiasis	16 (8.1%)	5 (17.9%)	21 (9.3%)	0.154	0.353
Missing values	24	4	28		
HIV	1 (0.5%)	2 (7.1%)	3 (1.3%)	**0.042***	0.159
Missing values	24	4	28		
Severe malaria	28 (12.7%)	2 (6.3%)	30 (11.9%)	0.391	0.521
Neurological malaria	3 (1.5%)	0 (0.0%)	3 (1.3%)	>0.999	>0.999
Missing values	24	4	28		
Severe anemia	57 (28.9%)	6 (21.4%)	63 (28.0%)	0.408	0.521
Missing values	24	4	28		
Other anemia	57 (28.9%)	6 (21.4%)	63 (28.0%)	0.408	0.521
Missing values	24	4	28		
Type of discharge				**<0.001*** **	<0.001
Recovered/Transferred	187 (85.4%)	2 (6.3%)	189 (75.3%)		
Absconded	32 (14.6%)	0 (0.0%)	32 (12.7%)		
Death	0 (0.0%)	30 (93.8%)	30 (12.0%)		
Missing values	2	0	2		
F100d frequency below target					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100d frequency above target				0.712	0.862
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 0.0	0 ; 1.0		
F100d quantity below target					

Characteristic	Non-response to treatment in Phase				
	FALSE N = 221[1]	**TRUE** N = 32[1]	**Overall** N = 253[1]	**p-value**[23]	**q-value**[4]
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100d quantity above target					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100d adherence below target					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100d adherence above target					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
Weight gain g/kg/day				**0.028***	0.130
Median (Q1 ; Q3)	19 (12.5 ; 26)	6 (4.8 ; 8)	13 (6.4 ; 22)		
Min ; Max	6 ; 29.0	4 ; 9.1	4 ; 29.0		
Missing values	213	28	241		
Persistent edema in day7	0 (0.0%)	1 (5.9%)	1 (1.1%)	0.185	0.386
Missing values	146	15	161		
Death	0 (0.0%)	30 (93.8%)	30 (12.0%)	**<0.001*** **	<0.001
Missing values	2	0	2		
Nutritional treatment				0.924	>0.999
F100d	50 (22.6%)	7 (21.9%)	57 (22.5%)		
Others	171 (77.4%)	25 (78.1%)	196 (77.5%)		

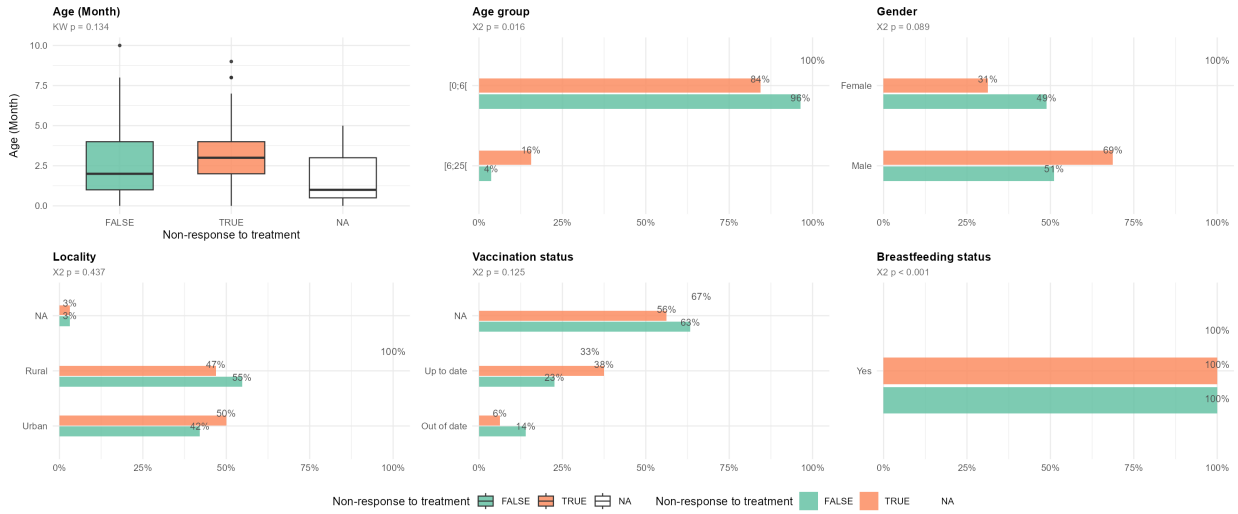
[1] n (%)

[2] Wilcoxon rank sum test; Fisher's exact test; Pearson's Chi-squared test; NA; Wilcoxon rank sum exact test

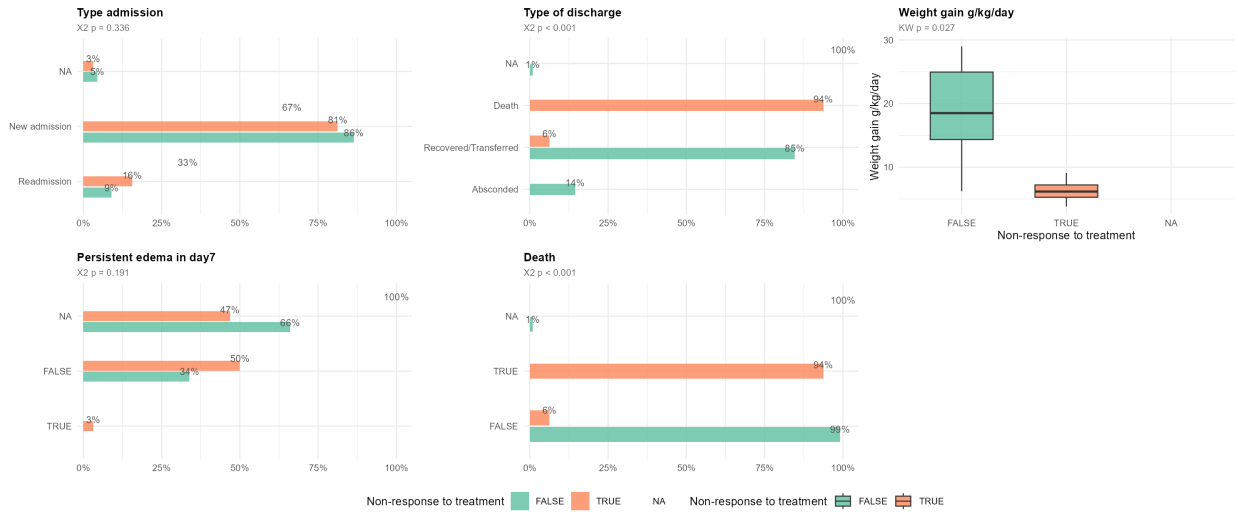
[3] *p<0.05; **p<0.01; ***p<0.001

[4] False discovery rate correction for multiple testing

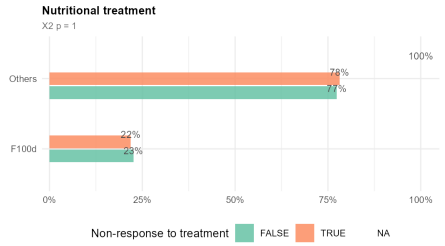
Sociodemographic



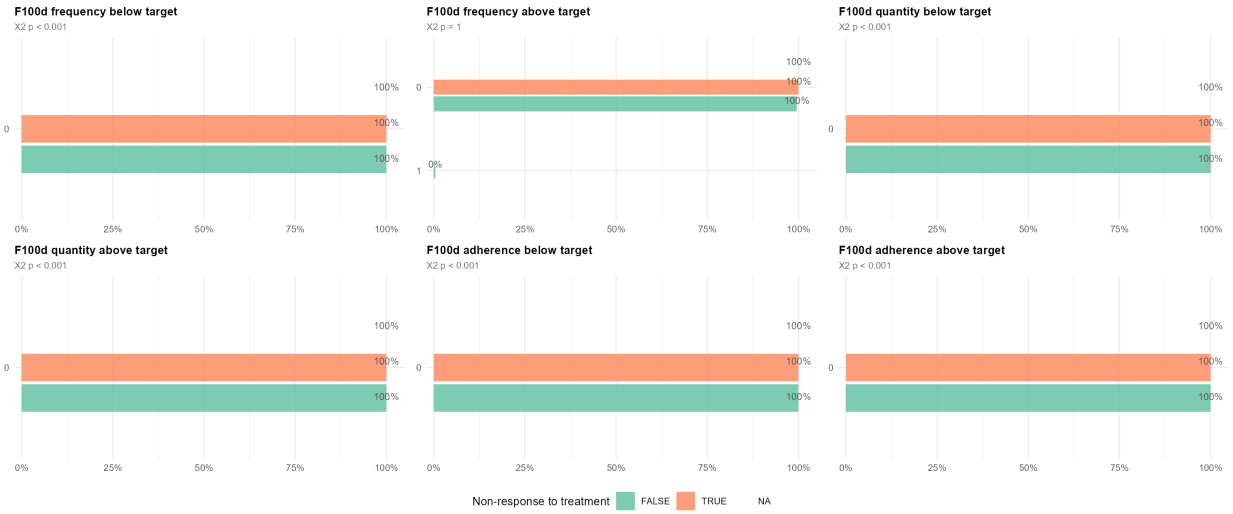
Clinical Admission and Severity



Nutritional Treatment

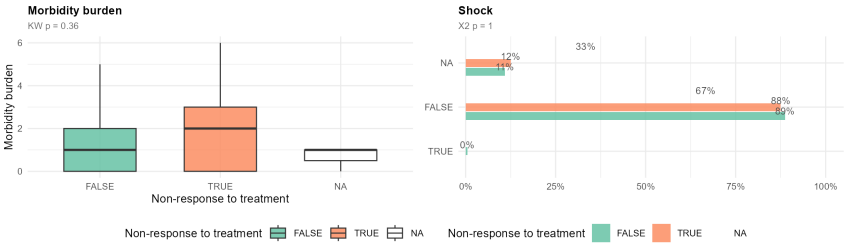


Feeding Compliance

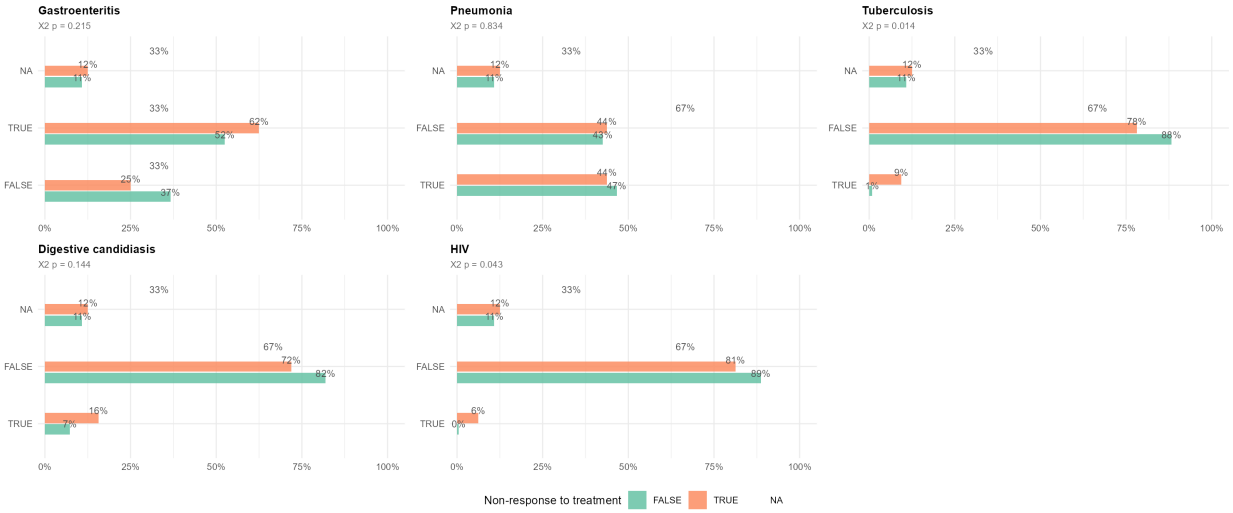


Morbidity and Comorbidities

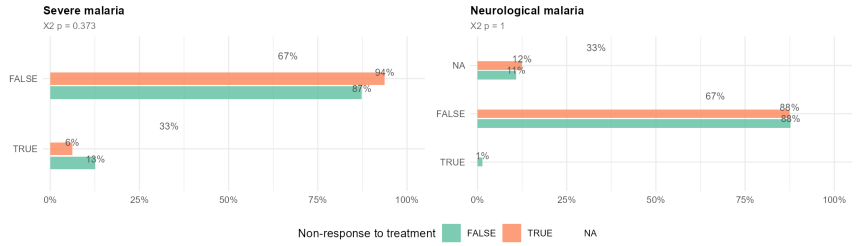
General



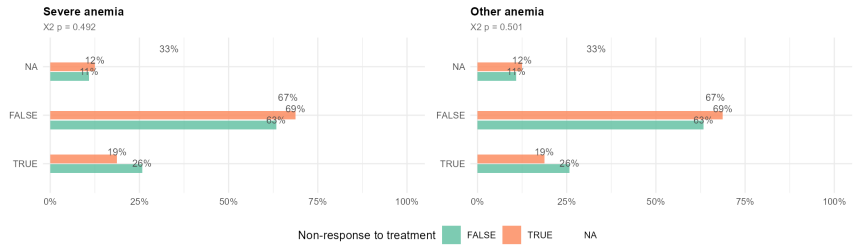
Infections and Parasitic



Malaria



Anaemia



Insert comments here.

Table 2: Results of the univariate logistic regression analysis between explanatory variables and outcome.

Predictor	Event N	OR	95% CI	p-value
Tuberculosis				**0.011**
FALSE	25	—	—	
TRUE	3	11.7	1.86, 92.1	
HIV				**0.027**
FALSE	26	—	—	
TRUE	2	15.1	1.40, 331	
Digestive candidiasis				0.13
FALSE	23	—	—	
TRUE	5	2.46	0.75, 6.97	
Gastroenteritis				0.2
FALSE	8	—	—	
TRUE	20	1.75	0.76, 4.39	
Severe malaria				0.3
FALSE	30	—	—	
TRUE	2	0.46	0.07, 1.64	
Type admission				0.3
New admission	26	—	—	
Readmission	5	1.84	0.57, 5.00	
Locality				0.4
Urban	16	—	—	
Rural	15	0.72	0.34, 1.54	
Severe anemia				0.4
FALSE	22	—	—	
TRUE	6	0.67	0.24, 1.65	
Shock				0.6
FALSE	28	—	—	
TRUE	0	0.00		
Pneumonia				0.8
FALSE	14	—	—	
TRUE	14	0.91	0.41, 2.03	

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Insert comments here.

5. Multivariate logistic regression analyses

Table 3 : Results of the generalize logistic regression analysis at Phase 1.

Predictor	**N**	**Event N**	**OR**[1]	**95% CI**	**p-value**	**q-value**[2]	**VIF**
Tuberculosis	207		*		0.034	**0.069**	1.0
FALSE		23		—			
TRUE		3	8.86	1.19, 74.3			
HIV	207				0.093	**0.093**	1.0
FALSE		24		—			
TRUE		2	9.67	0.67, 233			

[1]*p<0.05; **p<0.01; ***p<0.001

[2]False discovery rate correction for multiple testing

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, VIF = Variance Inflation Factor

Insert comments here.